Foundations of Reinforcement Learning and Interactive Decision Making – Exercises Solutions

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Chapter 1

Introduction

Exercise 1.1 Proposition 1, Part 2.

Consider the setting of Proposition 1 where $(x^1, y^1), \ldots, (x^T, y^T)$ are i.i.d., $\mathcal{F} = \{f : \mathcal{X} \to [0,1]\}$ is finite, the true regression function satisfies $f^* \in \mathcal{F}$, and $Y_i \in [0,1]$ almost surely. Prove that empirical risk minimizer $\widehat{f}$ with respect to square loss satisfies the following bound on excess risk. With probability at least $1-\delta$:

$$\mathcal{E}(\widehat{f}) \preceq \frac{\log(|\mathcal{F}|/\delta)}{T}.$$

1. For a fixed function $f \in \mathcal{F}$, consider the random variable

$$Z_i(f) = (f(x^i) - y^i)^2 - (f^*(x^i) - y^i)^2$$

for $i = 1, \ldots, T$. Show that

$$\mathbb{E}\left[Z_i(f)\right] = \mathbb{E}\left[(f(x^i) - f^*(x^i))^2\right] = \mathcal{E}(f).$$

2. Show that for any fixed $f \in \mathcal{F}$, the variance $\mathbb{V}\left[Zi(f)\right]$ is bounded as

$$\mathbb{V}\left[Z_i(f)\right] \leq 4\mathbb{E}\left[(f(x^i) - f^*(x^i))^2\right].$$

3. Apply Bernstein's inequality (Lemma 5) to show that with for any $f \in \mathcal{F}$, with probability at least $1-\delta$,

$$\mathcal{E}(f) \leq 2(\widehat{L}(f) - \widehat{L}(f^*)) + \frac{C\log(1/\delta)}{T},$$

for an absolute constant C, where $\widehat{L}(f) = \frac{1}{T}\sum_{t=1}^{T}(f(x^t) - y^t)^2$.

4. Extend this probabilistic inequality to simultaneously hold for all $f \in \mathcal{F}$ by taking the union bound over $f \in \mathcal{F}$. Conclude as a consequence that the bound holds for $\widehat{f}$, the empirical minimizer, implying (1.38)

Solution : 1. Developping the squares and rearranging the terms, we have

$$Z_i(f) = (f(x^i) - f^*(x^i))^2 + 2(f^*(x^i) - y_i)(f^*(x^i) - f(x^i)).$$

Now remark that $\mathbb{E}\left[f^*(x_i) - y_i)\right] = \mathbb{E}\left[\mathbb{E}\left[f^*(x_i) - y_i\right|x_i\right]] = 0$ where we have used the tower rule for the first equality and the fact that $f^*(x_i) = \mathbb{E}\left[y_i\left|x = x_i\right.\right]$. Thus

$$\begin{aligned}\mathbb{E}\left[Z_i(f)\right] &= \mathbb{E}\left[(f(x^i) - f^*(x^i))^2\right] + \mathbb{E}\left[2(f^*(x^i) - y_i)(f^*(x^i) - f(x^i))\right] \text{ (by linearity of } \mathcal{E}.) \\ &= \mathbb{E}\left[(f(x^i) - f^*(x^i))^2\right]\end{aligned}$$

Finally the fact that $\mathbb{E}\left[(f(x^i) - f^*(x^i))^2\right] = \mathcal{E}(f)$ comes from Lemma 1.

2. From König-Huygens formula we can write

$$\mathbb{V}\left[Z_i(f)\right] = \mathbb{E}\left[Z_i(f)^2\right] - \mathbb{E}\left[Z_i(f)\right]^2.$$

By definition $f : \mathcal{X} \to [0, 1]$.

3. First we recall Bernstein's inequality:

Lemma 1. *Let $Z_1, \ldots, Z_T$ be i.i.d with variance $\mathbb{V}\left[Z_i(f)\right] = \sigma^2$, and range $Z - \mathbb{E}\left[Z\right] \leq B$ almost surely. Then with probability at least $1 - \delta$,*

$$\frac{1}{T}\sum_{i=1}^{T} Z_i - \mathbb{E}\left[Z\right] \leq \sigma\sqrt{\frac{2\log(1/\delta)}{T}} + \frac{B\log(1/\delta)}{3T}.$$

Applying Lemma 1 for $Z_i = -Z_i(f)$ where $B = 2$ and $\sigma = 2\sqrt{\mathcal{E}(f)}$ yields

$$\widehat{L}(f^*) - \widehat{L}(f) + \mathcal{E}(f) \leq \sqrt{\frac{8\mathcal{E}(f)\log(1/\delta)}{T}} + \frac{2\log(1/\delta)}{3T} \tag{1.1}$$

where we have used the fact that $\widehat{L}(f) - \widehat{L}(f^*) = \frac{1}{T}\sum_{t=1}^{T} Z_i(f)$. Now in order to get the bound expected in the question we will need to use the following lemma which is a rewriting of the AMGM inequality.

Lemma 2. *For any $a, b \geq 0$ and $\eta > 0$ we have,*

$$\sqrt{ab} \leq \frac{a}{4\eta} + \eta b.$$

Proof. Starting from AMGM we have $\sqrt{xy} \leq \frac{x+y}{2}$. Since the inequality holds for any $x, y \geq 0$ we can pick $x = \frac{a}{2\eta}$ and $y = 2\eta b$. to conclude the proof. □

Let's apply Lemma 2 with $a = 8\mathcal{E}(f)$ and $b = \frac{\log(1/\delta)}{T}$, then for $\eta > 0$ we have

$$\sqrt{\frac{8\mathcal{E}(f)\log(1/\delta)}{T}} \leq \frac{2\mathcal{E}(f)}{\eta} + \eta\frac{\log(1/\delta)}{T}.$$

Taking $\eta = \frac{1}{2}$, substituing the above inequality in 1.1 and rearranging terms gives

$$\mathcal{E}(f) \leq 2(\widehat{L}(f^*) - \widehat{L}(f)) + \frac{7\log(1/\delta)}{6T}.$$

4. For sake of concreteness we detail the union bound argument. Let $\delta' > 0$, from the previous question we have

$$\begin{aligned}
\mathbb{P}\left[\exists f \in \mathcal{F} : \mathcal{E}(f) \geq 2(\widehat{L}(f^*) - \widehat{L}(f)) + \frac{7\log(1/\delta')}{6T}\right] &= \mathbb{P}\left[\bigcup_{f\in\mathcal{F}} \mathcal{E}(f) \geq 2(\widehat{L}(f^*) - \widehat{L}(f)) + \frac{7\log(1/\delta')}{6T}\right] \\
&\leq \sum_{f\in\mathcal{F}} \mathbb{P}\left[\mathcal{E}(f) \geq 2(\widehat{L}(f^*) - \widehat{L}(f)) + \frac{7\log(1/\delta')}{6T}\right] \\
&\leq \delta'|\mathcal{F}|
\end{aligned}$$

Now taking $\delta' = \frac{\delta}{|\mathcal{F}|}$ we have with probability $1-\delta$ that for any f

$$\mathcal{E}(f) \leq 2(\widehat{L}(f^*) - \widehat{L}(f)) + \frac{7\log(|\mathcal{F}|/\delta)}{6T}$$

and in particular this holds for $\widehat{f}$. ◻

Exercise 1.2 ERM in Online Learning

Consider the problem of Online Supervised Learning with indicator loss $\ell(f(x), y) = \mathbb{I}\{f(x) \neq y\}$, $\mathcal{Y} = \mathcal{Y}' = \{0, 1\}$, and a finite class $\mathcal{F}$.

1. Exhibit a class $\mathcal{F}$ for which ERM cannot ensure sublinear growth of regret for all sequences, i.e. there exists a sequence $(x^1, y^1), \ldots, (x^T, y^T)$ such that

$$\sum_{t=1}^{T} \ell(\hat{f}^t(x^t), y^t) - \min_{f\in\mathcal{F}} \sum_{t=1}^{T} \ell(f(x^t), y^t) = \Omega(T),$$

   where $\hat{f}^t$ is the empirical minimizer for the indicator loss on $(x^1, y^1), \ldots, (x^{t-1}, y^{t-1})$. Note : The construction must have $|\mathcal{F}| \leq C$, where C is an absolute constant that does not depend on T.

2. Show that if data are i.i.d., then in expectation over the data, ERM attains a sublinear bound $O(\sqrt{T \log |\mathcal{F}|})$ on regret for any finite class $\mathcal{F}$.

Solution : 1. We will do a constrution with $|\mathcal{F}| = 2$, we consider the two constants functions $f_0 = 0$ and $f_1 = 1$. We look at the following $(y_t)_{t\geq 1}$:

$$1, 0, 0, 1, 1, 0, 0, \ldots$$

Everytime the algorithm sees more 1 than 0, it will predict 1 and make a mistake and inversely it it sees more 0 than 1 (If there is the same number, ties are broken arbitrarilies and the algorithm will be wrong half the time). That means that after T round, our algorithm makes at least $\frac{3T}{4} - 3$ mistakes. Conversely, either f_1 or f_0 will make less than $\frac{T}{2}$ mistakes. This means that ERM suffers linear regret on that sequence.

2. We operate under the more general assumption that ℓ is bounded in $[0, 1]$. We introduce some notation :

- For a fixed $f \in \mathcal{F}$, $\hat{\ell}_t(f) = \sum_{s=1}^{t} \ell(f(x_t), y_t)$, $\hat{f}^{t+1} = \arg\min_{f \in \mathcal{F}} \hat{\ell}_t(f)$.
- $\mu \in \Delta(\mathcal{X} \times \mathcal{Y})$ the distribution under which the data is drawn.
- For a fixed $f \in F$, $\bar{\ell}(f) = \mathbb{E}_{x,y \sim \mu}[\ell(f(x), y)]$, $f^* = \arg\min_{f \in \mathcal{F}} \bar{\ell}(f)$.

We start by a decomposition of the regret :

$$\begin{aligned}
\mathbb{E}[R_n] &= \mathbb{E}\left[\sum_{t=1}^{T} \ell(\hat{f}^t(x_t), y_t) - \min_{f \in \mathcal{F}} \sum_{t=1}^{T} \ell(f(x_t), y_t)\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{T} \left[\ell(\hat{f}^t(x_t, y_t)) - \bar{\ell}(f^*)\right]\right] + T\mathbb{E}\left[\bar{\ell}(f^*) - \hat{\ell}_T(\hat{f}^{T+1})\right]
\end{aligned}$$

Our main technical tool is Hoeffding's Lemma. Since ℓ is bounded, we have that for a fixed f, for any $\delta > 0$

$$\mathbb{P}\left[|\bar{\ell}(f) - \hat{\ell}_t(f)| \geq \sqrt{\frac{\log\left(\frac{2}{\delta}\right)}{2t}}\right] \leq \delta.$$

In particular, taking a union bound over all functions in $\mathcal{F}$, we have that :

$$\mathbb{P}\left[\exists f \in \mathcal{F}, |\bar{\ell}(f) - \hat{\ell}_t(f)| \geq \sqrt{\frac{\log\left(\frac{2|\mathcal{F}|}{\delta}\right)}{2t}}\right] \leq \delta.$$

We also state the following Lemma for integrating tail bounds :

Lemma 3. *Let A,B be positive reals and X a real valued random variable such that $\mathbb{P}\left[X \geq \sqrt{A \log\left(\frac{B}{\delta}\right)}\right] \leq \delta$, then*

$$\mathbb{E}[X] \leq 2\sqrt{A \log B}.$$

Proof. We have by a change of variable $\mathbb{P}\left[X - \sqrt{A\log(B)} \geq \epsilon^2\right] \leq \exp\left(\frac{-\epsilon^2}{A}\right)$. Then

$$\begin{aligned}
\mathbb{E}\left[X\right] &= \sqrt{A\log B} + \mathbb{E}\left[X - \sqrt{A\log B}\right] \\
&\leq \sqrt{A\log B} + \int_0^\infty \mathbb{P}\left[X - \sqrt{A\log B} \geq \epsilon\right] d\epsilon \\
&= \sqrt{A\log B} + \int_0^\infty \exp -\frac{\epsilon^2}{A}\, d\epsilon \\
&= \sqrt{A\log B} + \frac{\sqrt{A\pi}}{2} \\
&\leq 2\sqrt{A\log B}.
\end{aligned}$$

□

Now, with probability at least $1-\delta$, we have

$$|\bar{\ell}(f^*) - \hat{\ell}_T(f^{T+1})| \leq \sqrt{\frac{\log\left(\frac{2|\mathcal{F}|}{\delta}\right)}{2T}}.$$

Integrating this bound with Lemma 3, we get

$$T\mathbb{E}\left[\bar{\ell}(f^*) - \hat{\ell_T}(f^{\hat{T}+1})\right] \leq \sqrt{2T\log(2|\mathcal{F}|)}.$$

It remains to deal with the term $\sum_{t=1}^T \ell(\hat{f}^t(x_t, y_t)) - \bar{\ell}(f^*)$. Here, since (x_t, y_t) is independent from $\hat{f}^t$, we have that for any $t \geq 2$,

$$\mathbb{E}\left[\ell(\hat{f}^t(x_t), y_t)\,|(x_1, y_1, \ldots, x_{t-1}, y_{t-1})\right] = \bar{\ell}(\hat{f}^t)$$

Now we bound with probability at least $1-\delta$

$$\begin{aligned}
\bar{\ell}(\hat{f}^t) &= \hat{\ell}_{t-1}(\hat{f}^t) + (\bar{\ell}(\hat{f}^t) - \hat{\ell}_{t-1}(\hat{f}^t)) \\
&\leq \hat{\ell}_{t-1}(f^*) + (\bar{\ell}(\hat{f}^t) - \hat{\ell}_{t-1}(\hat{f}^t)) \\
&= \bar{\ell}(f^*) + (\hat{\ell}_{t-1}(f^*) - \bar{\ell}(f^*)) + (\bar{\ell}(\hat{f}^t) - \hat{\ell}_{t-1}(\hat{f}^t)) \\
&\leq \bar{\ell}(f^*) + \sqrt{\frac{2\log\left(\frac{2|\mathcal{F}|}{\delta}\right)}{t-1}}.
\end{aligned}$$

By the tower rule of expectation, we have that

$$\begin{aligned}
\mathbb{E}\left[\ell(\hat{f}^t(x_t), y_t) - \bar{\ell}(f^*)\right] &= \mathbb{E}\left[\mathbb{E}\left[\ell(\hat{f}^t(x_t), y_t)\,|(x_1, y_1, \ldots, x_{t-1}, y_{t-1})\right] - \bar{\ell}(f^*)\right] \\
&= \mathbb{E}\left[\bar{\ell}(\hat{f}^t) - \bar{\ell}(f^*)\right] \\
&\leq \sqrt{\frac{8\log\left(2|\mathcal{F}|\right)}{t-1}}.
\end{aligned}$$

Where the last line is obtained integrating the tail bound via Lemma 3. Now it only remains to put everything together, we have

$$\begin{aligned}
\mathbb{E}\left[R_n\right] &= \mathbb{E}\left[\sum_{t=1}^{T}\left[\ell(\hat{f}^t(x_t,y_t)) - \bar{\ell}(f^*)\right]\right] + T\mathbb{E}\left[\bar{\ell}(f^*) - \hat{\ell}_T(\hat{f}^{T+1})\right] \\
&\leq 1 + \sum_{t=2}^{T}\sqrt{\frac{8\log(2|\mathcal{F}|)}{t-1}} + \sqrt{2T\log(2|\mathcal{F}|)}. \\
&\leq \sqrt{32T\log(2|\mathcal{F}|)} + \sqrt{2T\log(2|F|)} \\
&= O(\sqrt{T\log|\mathcal{F}|}),
\end{aligned}$$

where the second inequality comes from the elementary inequality $\sum_{t=2}^{T}\frac{1}{\sqrt{t-1}} \leq 2(\sqrt{T}-1)$. $\square$

Exercise 1.3 Low Noise

1. For a nonnegative random variable X, prove that for any $\eta > 0$,

$$\log\mathbb{E}\left[\exp\left(-\eta(X-\mathbb{E}\left[X\right])\right)\right] \leq \frac{\eta^2}{2}\mathbb{E}\left[X^2\right].$$

Hint : use the fact that $\log x \leq x-1$ and $\exp(-x) \leq 1-x+x^2/2$for $x \geq 0$.

2. Consider the setting of Proposition 3, Part 1(Generic Loss). Prove that the randomized variant of the Exponential Weights Algorithm satisfies, for any $f^* \in \mathcal{F}$,

$$\sum_{t=1}^{T}\mathbb{E}_{\hat{f}^t\sim q^t}\left[\ell(\hat{f}^t(x^t),y^t)\right] - \ell(f^*(x^t),y^t) \leq \frac{\eta}{2}\sum_{t=1}^{T}\mathbb{E}_{\hat{f}^t\sim q^t}\left[\ell(\hat{f}^t(x^t),y^t)^2\right] + \frac{\log|\mathcal{F}|}{\eta}.$$

for any sequence of data and nonnegative losses. Hint : replace Hoeffding's Lemma by the result of the first question.

3. Suppose $\ell(f(x),y) \in [0,1]$ for all $x \in \mathcal{X}, y \in \mathcal{Y}$ and $f \in \mathcal{F}$. Suppose that there is a "perfect expert $f^* \in \mathcal{F}$ such that $\ell(f^*(x^t)y^t) = 0$ for all $t \in [T]$. Conclude that the above algorithm, with an appropriate choice of η, enjoys a bound of $O(\log|\mathcal{F}|)$ on the cumulative loss of the algorithm (equivalently, the fast rate $\frac{\log|F|}{T}$ for the average regret). This setting is called "zero noise."

4.Consider the binary classification problem with indicator loss, and suppose $\mathcal{F}$ contains a perfect expert, as above. The *Halving Algorithm* maintains a version space $\mathcal{F}^t = \{f \in \mathcal{F} : f(x^s) = y^s, s < t\}$ and given x^t, follows the majority vote of remaining experts in $\mathcal{F}^t$. Show that this algorithm incurs cumulative loss at most $O(\log|\mathcal{F}|)$. Hence, the Exponential Weights

Algorithm can be viewed as an extension of the Halving algorithm to settings where the optimal loss is non-zero.

Solution : 1. Using the hint, we have

$$\begin{aligned}
\log \mathbb{E}\left[\exp\left(-\eta(X-\mathbb{E}[X])\right)\right] &= \eta\mathbb{E}[X] + \log \mathbb{E}[\exp -\eta X] \\
&\leq \eta\mathbb{E}[X] + \mathbb{E}[\exp(-\eta X) - 1] \\
&\leq \eta\mathbb{E}[X] + \mathbb{E}\left[1 - \eta X + \frac{\eta^2 X^2}{2} - 1\right] \\
&= \eta\mathbb{E}[X] - \eta\mathbb{E}[X] + \frac{\eta^2}{2}\mathbb{E}\left[X^2\right] \\
&= \frac{\eta^2}{2}\mathbb{E}\left[X^2\right].
\end{aligned}$$

Where the first inequality is valid because the exponential is always positive and the second one because $\eta X \geq 0$.

2. We take the same notation as in the proof of Proposition 3. We have the potential $\Phi_\eta^t = -\log \sum_{f\in\mathcal{F}} \exp\left(-\eta \sum_{i=1}^t \ell(f(x^i), y^i)\right)$. Rewriting the inequality of question 1 as $\eta\mathbb{E}[X] \leq -\log \mathbb{E}[\exp X] + \frac{\eta^2}{2}\mathbb{E}\left[X^2\right]$ and applying it to $X = \ell(\hat{f}^t(x^t), y^t) \geq 0$ where $\hat{f}^t \sim q_t$, we get

$$\eta\mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)\right] \leq -\log \mathbb{E}_{f\sim q_t}\left[\exp\left(\ell(f(x^t), y^t)\right)\right] + \frac{\eta^2}{2}\mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)^2\right].$$

As in the proof of Proposition 3, we remark that

$$-\log \mathbb{E}_{f\sim q_t}\left[\exp\left(\ell(f(x^t), y^t)\right)\right] = \Phi_\eta^t - \Phi_\eta^{t-1},$$

and summing over t we get

$$\eta \sum_{t=1}^T \mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)\right] \leq \Phi_h^T - \Phi_n^0 + \frac{\eta^2}{2}\sum_{t=1}^T \mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)^2\right].$$

As in the proof of Proposition 3, we can upper bound for any $f^* \in \mathcal{F}$.

$$\Phi_\eta^T \leq \eta \sum_{t=1}^T \ell(f^*(x^t), y^t),$$

while $\Phi_\eta^0 = -\log|\mathcal{F}|$. Hence we have that for any $f^* \in \mathcal{F}$

$$\sum_{t=1}^T \mathbb{E}_{\hat{f}^t\sim q^t}\left[\ell(\hat{f}^t(x^t), y^t)\right] - \ell(f^*(x^t), y^t) \leq \frac{\eta}{2}\sum_{t=1}^T \mathbb{E}_{\hat{f}^t\sim q^t}\left[\ell(\hat{f}^t(x^t), y^t)^2\right] + \frac{\log|\mathcal{F}|}{\eta}.$$

3. We choose $\eta = 1$, using the fact that $\ell(\cdot,\cdot) \in [0,1]$, we have that $\ell^2 \leq \ell$. Hence with the result of the previous question and the choice of f^* as the perfect expert, we get

$$\sum_{t=1}^{T} \mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)\right] \leq \frac{1}{2}\sum_{t=1}^{T} \mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)\right] + \log|\mathcal{F}|.$$

And

$$\sum_{t=1}^{T} \mathbb{E}_{f\sim q_t}\left[\ell(f(x^t), y^t)\right] - \ell(f^*(x_t), y_t) \leq 2\log|\mathcal{F}|.$$

4. It is easy to see that for this algorithm at a given round, either no mistakes is made or at least half of the experts are discarded. Since this can happen only $\lfloor \log_2|\mathcal{F}| \rfloor$ times, the regret is of order $O(\log|\mathcal{F}|)$.

$\square$

Chapter 2

Multi-Armed Bandits

Chapter 3

Contextual Bandits

Chapter 4

Structured Bandits

Chapter 6

General Decision Making